



The Challenge

As awareness of indoor air quality grew, schools and public spaces increasingly needed simple, real-time indicators to identify when ventilation was inadequate. traditional HVAC systems typically operate on fixed schedules rather than adapting continuously to actual occupancy or CO₂ buildup, creating potential health and safety gaps.

Key design challenges included:

- Non-technical usability** — Users had to immediately understand device indicators without any training
- Intuitive setup** — Educators and administrators had to be able to complete device configuration independently
- Scalable software** — Mobile and desktop interfaces had to work seamlessly at the district, school, and room levels.
- System integration** — The solution required building system connectivity (HVAC, BMS) for automated response capabilities

